2.1.3. Introduction Double Bonds:

• The introduction of double bonds may cause two main effects

A. The possibility of the existence of geometrical isomery and thus different activity.
- e.g. unsaturated propyl amine H_1-antagonists.



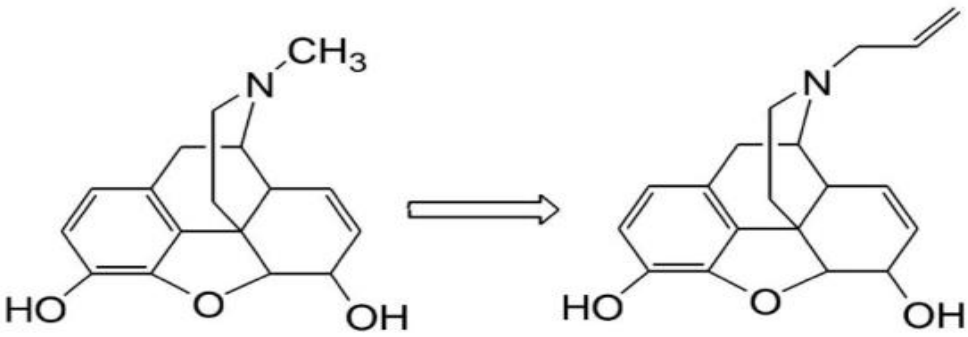
E-isomer
(>1000 time the affinity of Z)



Z-isomer

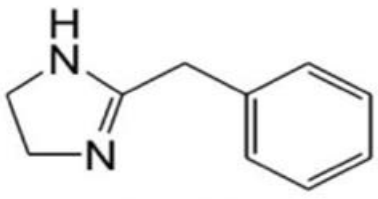
B. Altering physicochemical properties, it can modify biological activity

example is the conversion of morphinic receptor agonists into the corresponding antagonists in case of morphine.

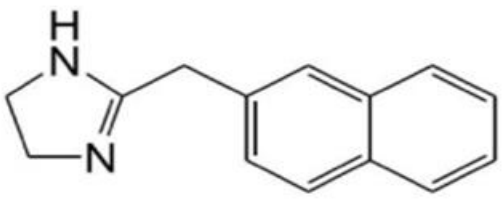


Morphine Nalorphine

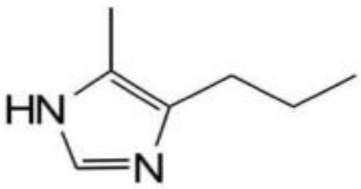
2.1.4. Introduction, Removal, or Replacement of Bulky groups:



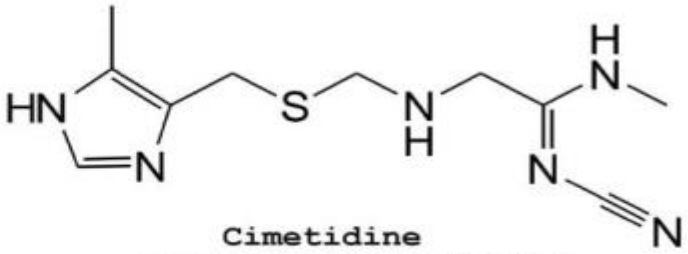
Tolazoline
α–Adrenergic antagonist



Naphazoline
α–Adrenergic agonist



4-Methylhistamine
H_2-receptor agonist



Cimetidine
H_2-receptor antagonist

2.2. Alterations of physical and chemical properties through introduction of new groups or the replacement of certain moieties by different ones,

this can be achieved through the following:

2.2.1. Isosteric Replacements:

Isosteres: are molecules or groups of atoms that have the same number and arrangements of electrons.

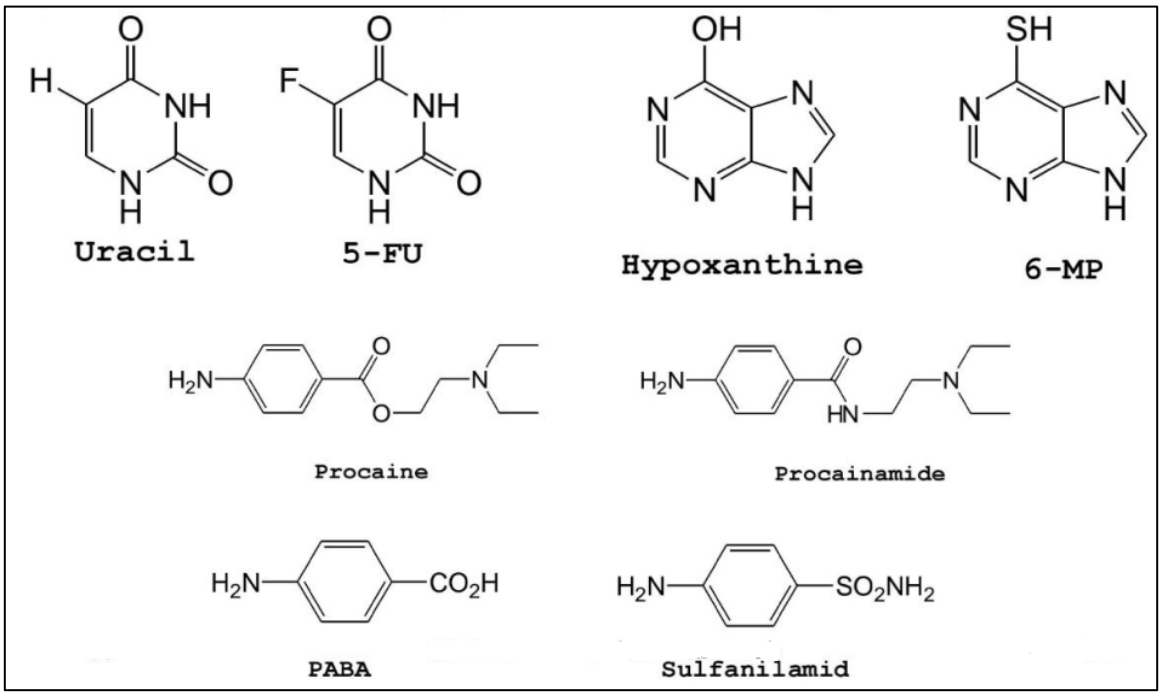
For example, N_2 and *CO*, N_3^- and NCO^-. These substances have similar physical properties

Bioisosterism or nonclassical isosteres: are groups or molecules which have chemical and physical similarities producing broadly similar biological properties.

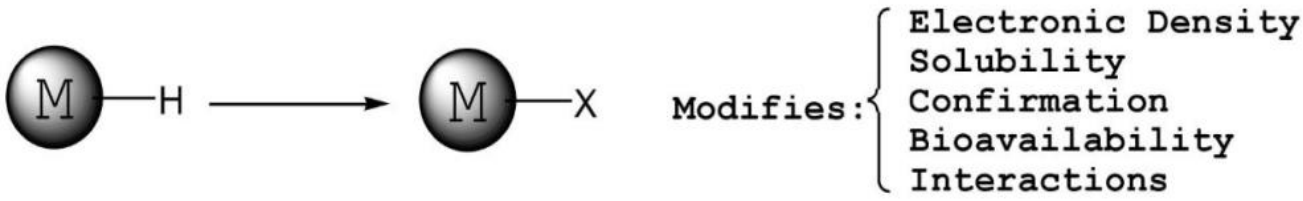
a- Classical Isosteres: they are divided into five classes as illustrated in the following table:

Class 1 (monovalent)	2 (divalent)	3 (trivalent)	4 (tetravalent)	5 (rings)
F,Cl,Br,I OH,SH NH_2, PH_2 CH_3	-O- -S- -Se- -Te-	-N= -P= -As= -Sb= -CH=	=C= =Si= $=N^+=$ =P= =As= $=Sb^+=$	-CH=CH -S- -O- -NH-    

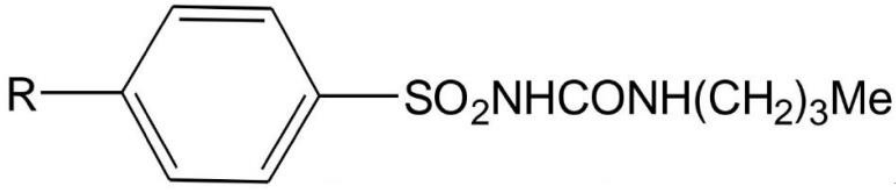
b- Nonclassical Isosteres: they are also known and include paired examples such as H and F, -CO_2H and -SO_3H, and -CO- and -SO_2-. Some of the examples of Isosteric replacement that have provided useful drugs are include:



2.2.2. Modifications toward Inhibition or promotion of Various Electronic states:



The half-life of tolbutamide (R=CH_3) can be significantly increased by replacement of methyl group by chlorine as in chlorpropamide (R=Cl).



4. Testing drugs:

- Tests are required in order to find lead compounds and for drug optimization
- Tests can be in vivo or in vitro or ex vivo
- A combination of tests is often used in research programmes
- **Definitions:**

- ***In vitro:*** In an artificial environment, as in a test tube or culture media
- ***In vivo:*** In the living body, referring to tests conducted in living animals
- ***Ex vivo:*** Usually refers to doing the test on a tissue taken from a living organism.

4.1 in vitro Tests

- Tests not carried out on animals/humans
  - Target molecules (e.g. isolated enzymes or receptors)
  - Cells (e.g. cloned cells)
  - Tissues (e.g. muscle tissue)
  - Organs or Micro-organisms (for antibacterial agents)
- More suitable for routine testing



- Used in high throughput screening
- Measure the interaction of a drug with the target but not the ability of the drug to reach the target
- Results are easier to rationalize - less factors involved
- Does not demonstrate a physiological or clinical effect
- Does not identify possible side effects
- Does not identify effective prodrugs

4.1.1 Enzyme Inhibition Tests

- Identify competitive or non-competitive inhibition
- Strength of inhibition measured as IC50
- IC50 = concentration of inhibitor required to reduce enzyme activity by 50%

4.1.2 Testing with Receptors

- Not easy to isolate membrane bound receptors
- Carried out on whole cells, tissue cultures, or isolated organs

• ***Affinity:*** strength with which compounds bind to a receptor

• ***Efficacy:*** measure of maximum biochemical effect resulting from binding of a compound to a receptor.

• ***Potency:*** concentration of an agonist required to produce 50% of the maximum possible effect.

4.2 in vivo Tests



• Carried out on live animals or humans

• Measure an observed physiological effect

• Measure a drug's ability to interact with its target and its ability to reach that target

• Can identify possible side effects

• Rationalization may be difficult due to the number of factors involved

• Transgenic animals - genetically modified animals

• Drug potency - concentration of drug required to produce 50% of the maximum possible effect

• Therapeutic ratio/index - compares the dose level of a drug required to produce a desired effect in 50% of the test sample (ED_{50}) versus the dose level that is lethal to 50% of the sample (LD_{50})

5- Structure-Activity Relationships (SAR)

SAR: is a statement of the effect of structure change on biological activity within a congeneric series (a family) of compounds.

- Both the affinity of a drug for its receptor and its intrinsic activity are determined by its chemical structure.

AIM: Identify which functional groups are important for binding and/or activity

Several methods are presently used to study SAR:

• Alter, remove or mask a functional group

• Test the analogue for activity

• Conclusions depend on the method of testing
  in vitro: tests for binding interactions with target
  in vivo: tests for target binding interactions and/or pharmacokinetics

• If in vitro activity drops, it implies group is important for binding

• If in vivo activity unaffected, it implies group is not important

Notes on analogues:

• Modifications may disrupt binding by electronic / steric effects

• Easiest analogues to make are those made from lead compound

• Possible modifications may depend on other groups present

• Some analogues may have to be made by a full synthesis (e.g. replacing an aromatic ring with a cyclohexane ring)

• Allows identification of important groups involved in binding

•Allows identification of the pharmacophore

Pharmacophore

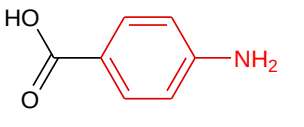
• Set of structural features in a drug molecule recognized by a receptor.
• The common denominator shared by a set of active molecules.

Importance:

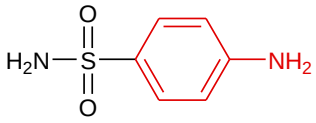
• Defines the important groups involved in binding **and** the relative positions of the binding groups

• Need to know Active Conformation

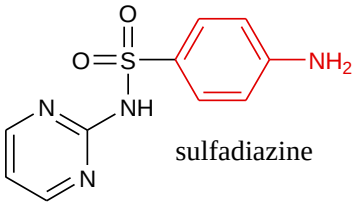
• Important to Drug Design and Drug Discovery



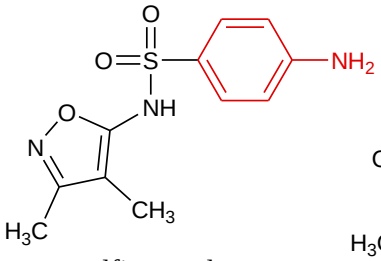
PABA



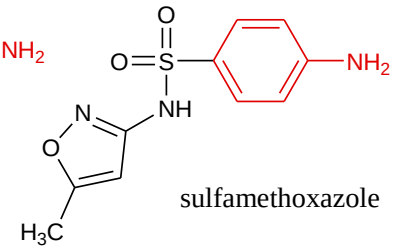
Sulfanilamide



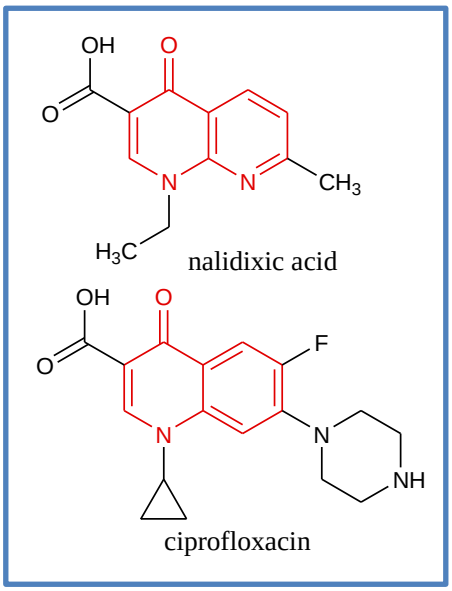
sulfadiazine



sulfisoxazole



sulfamethoxazole



Pharmacophoric descriptors are including:

a. H-bond sites
b. Hydrophobic and electrostatic interaction sites
c. Ring centers and virtual points
d. Distances, 3D relationship